Land portfolio: cost x return

Synthetic data generated by generate_data.py. Every number below is computed by the model in this repository.

Portfolio

10 parcels. VGV R$ 59.5 mi (subdivision R$ 52.2 mi, whole-lot resale R$ 7.3 mi).
Net profit R$ 24.0 mi, margin on VGV 40.4%.
NPV at 14% a year: R$ 2.3 mi. IRR 16.7% a year.
Maximum cash exposure R$ 28.6 mi in 2028-09; payback in 2031-02.

Scenarios

pessimistic: NPV R$ -11.9 mi, IRR 2.8%, max exposure R$ 40.1 mi
base: NPV R$ 2.3 mi, IRR 16.7%, max exposure R$ 28.6 mi
optimistic: NPV R$ 12.5 mi, IRR 31.7%, max exposure R$ 20.5 mi

What moves the result most

infra_per_m2: NPV from R$ 6.0 mi to R$ -3.8 mi (value 80 to 160)
lot_price_factor: NPV from R$ -2.9 mi to R$ 5.8 mi (value 0.85 to 1.1)
discount_rate_year: NPV from R$ 5.4 mi to R$ -1.0 mi (value 0.11 to 0.18)
absorption_months: NPV from R$ 4.4 mi to R$ 0.4 mi (value 24 to 48)

Subdivide or sell whole

T02 Área Jardim Novo: NPV subdivided R$ 0.4 mi vs sold whole R$ -0.6 mi
T04 Gleba Boa Vista: NPV subdivided R$ 0.3 mi vs sold whole R$ -0.6 mi
T05 Gleba Santa Rita: NPV subdivided R$ 0.5 mi vs sold whole R$ -0.3 mi
T09 Gleba Ribeirão Claro: NPV subdivided R$ 2.4 mi vs sold whole R$ -0.3 mi

Memos (AI-written, every number checked against the model)

T01: do_not_invest - T01 Sítio Pinheirinho does not clear the 14.0% hurdle rate, so the parcel should not be acquired on the current plan.

T02: invest_with_conditions - Parcel T02 Área Jardim Novo clears the 14.0% hurdle rate with an NPV of R$ 364,934, but the 56-month payback and R$ 4,31

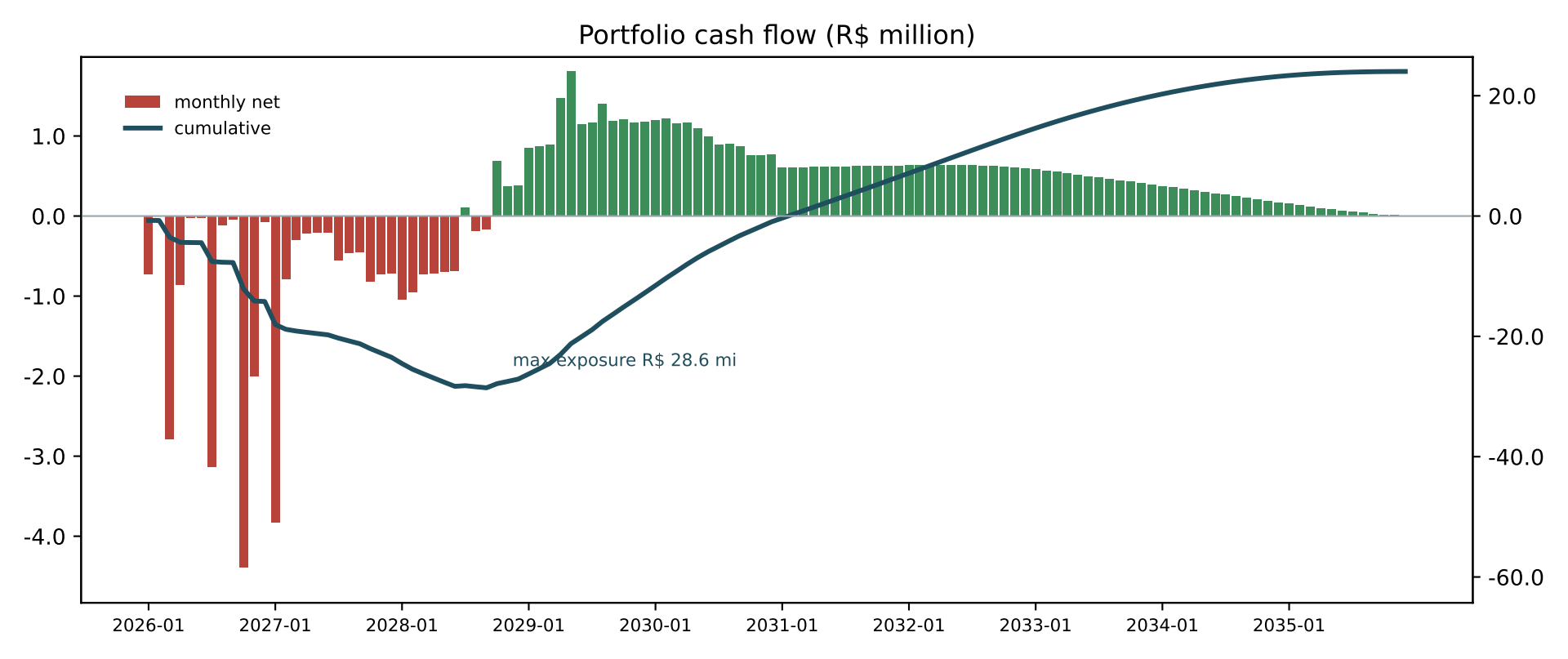
T03: do_not_invest - T03 Chácara Água Limpa does not clear the 14.0% hurdle rate, with an NPV of R$ -180,966 and an IRR of 5.2% per year.

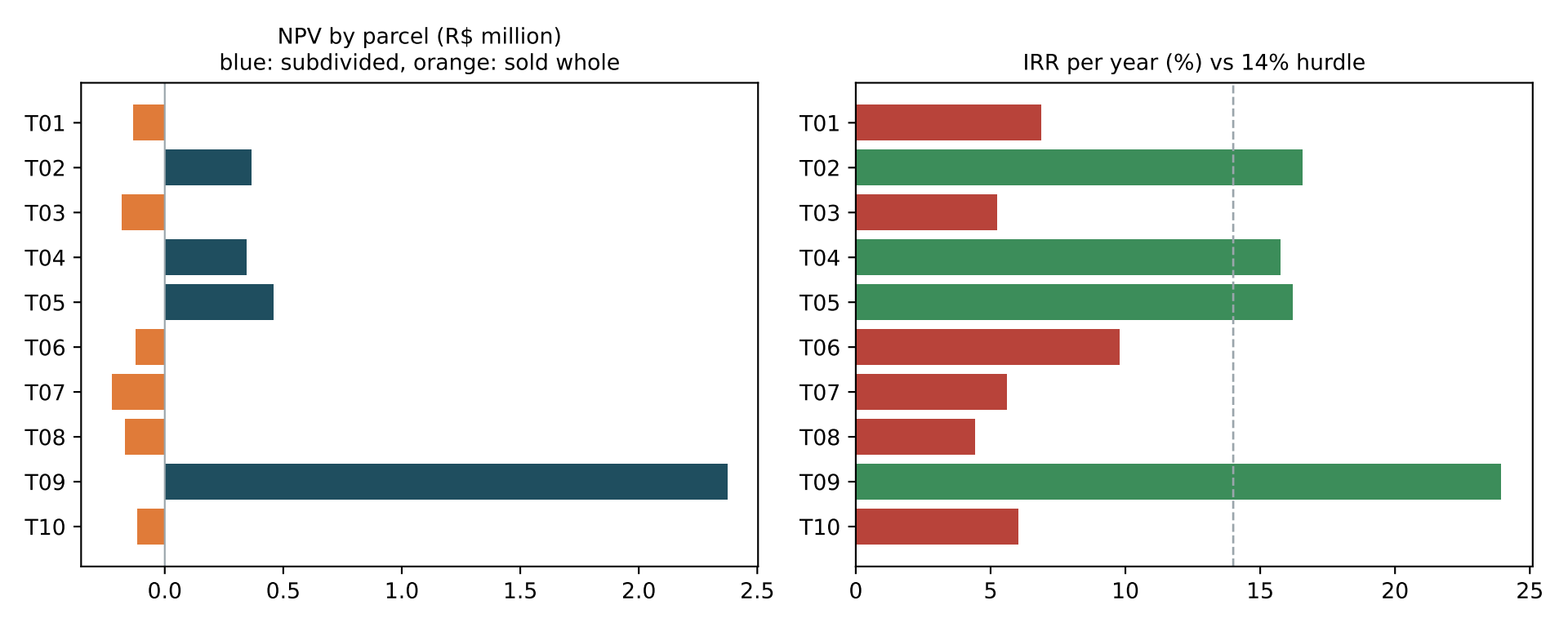
T04: invest_with_conditions - T04 Gleba Boa Vista, a 68,000 m2 subdivision expansion, clears the 14.0% hurdle rate with a positive NPV, but payback of

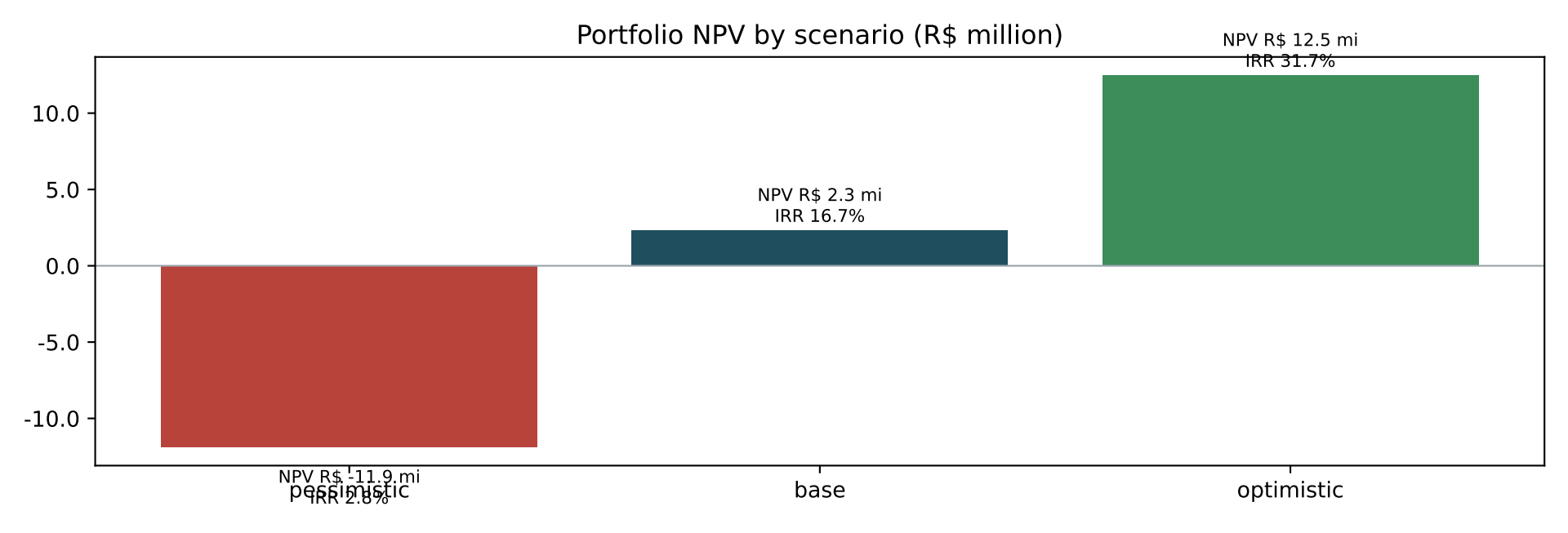
T05: invest_with_conditions - T05 Gleba Santa Rita, a 147-lot subdivision expansion, clears the 14.0% hurdle rate with positive NPV, but payback is sl

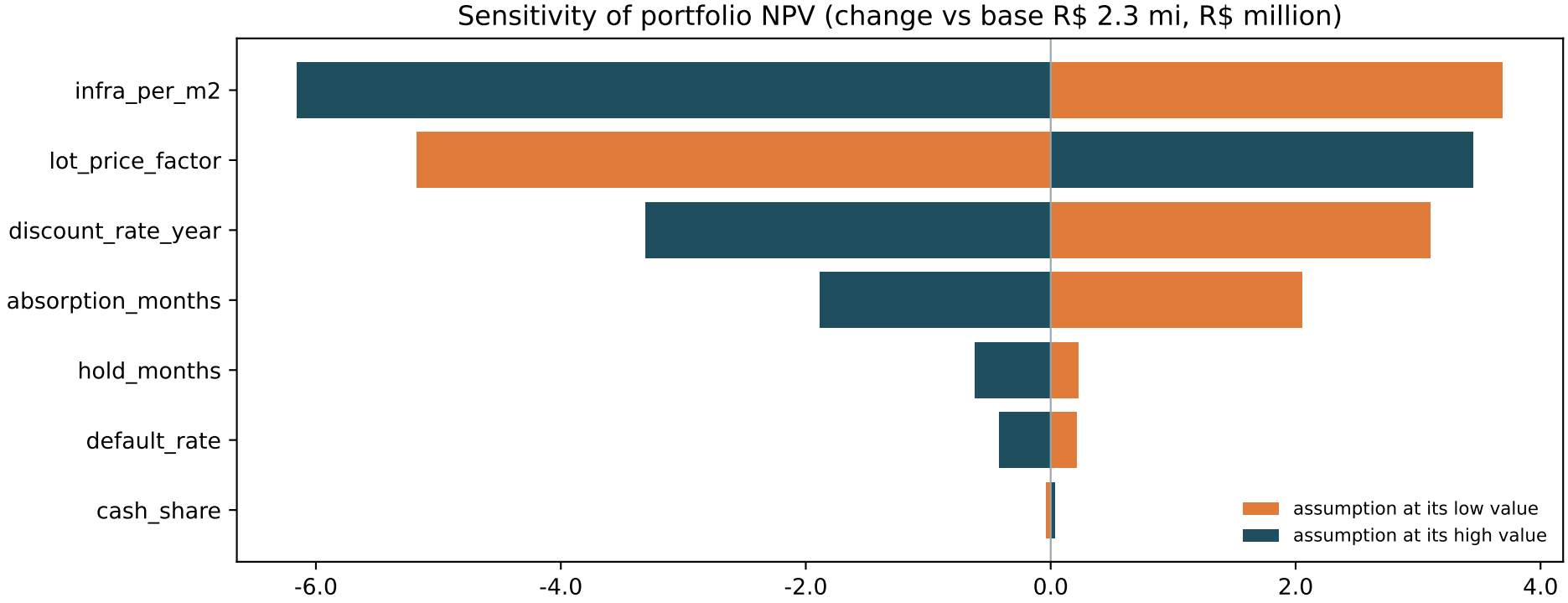
T06: do_not_invest - Parcel T06 Área Avenida Central fails the 14.0% hurdle rate, with a negative NPV of R$ -121,070 despite positive nominal profit.

T07: do_not_invest - T07 Terreno Esquina Comercial does not clear the 14.0% hurdle rate, with a negative NPV of R$ -223,173 at that rate.


Portfolio cash flow (R$ million)
monthly net
cumulative
max exposure R$ 28.6 mi
1.0
0.0
-1.0
-2.0
-3.0
-4.0
20.0
0.0
-20.0
-40.0
-60.0
2026-01
2027-01
2028-01
2029-01
2030-01
2031-01
2032-01
2033-01
2034-01
2035-01


NPV by parcel (R$ million)
blue: subdivided, orange: sold whole
T01
T02
T03
T04
T05
T06
T07
T08
T09
T10
0.0
0.5
1.0
1.5
2.0
2.5
IRR per year (%) vs 14% hurdle
T01
T02
T03
T04
T05
T06
T07
T08
T09
T10
0
5
10
15
20
25


Portfolio NPV by scenario (R$ million)
NPV R$ 12.5 mi
IRR 31.7%
NPV R$ 2.3 mi
IRR 16.7%
10.0
5.0
0.0
-5.0
-10.0
NPV R$ -11.9 mi
IRR 2.8%
pessimistic
base
optimistic


Sensitivity of portfolio NPV (change vs base R$ 2.3 mi, R$ million)
infra_per_m2
lot_price_factor
discount_rate_year
absorption_months
hold_months
default_rate
cash_share
assumption at its low value
assumption at its high value
-6.0
-4.0
-2.0
0.0
2.0
4.0